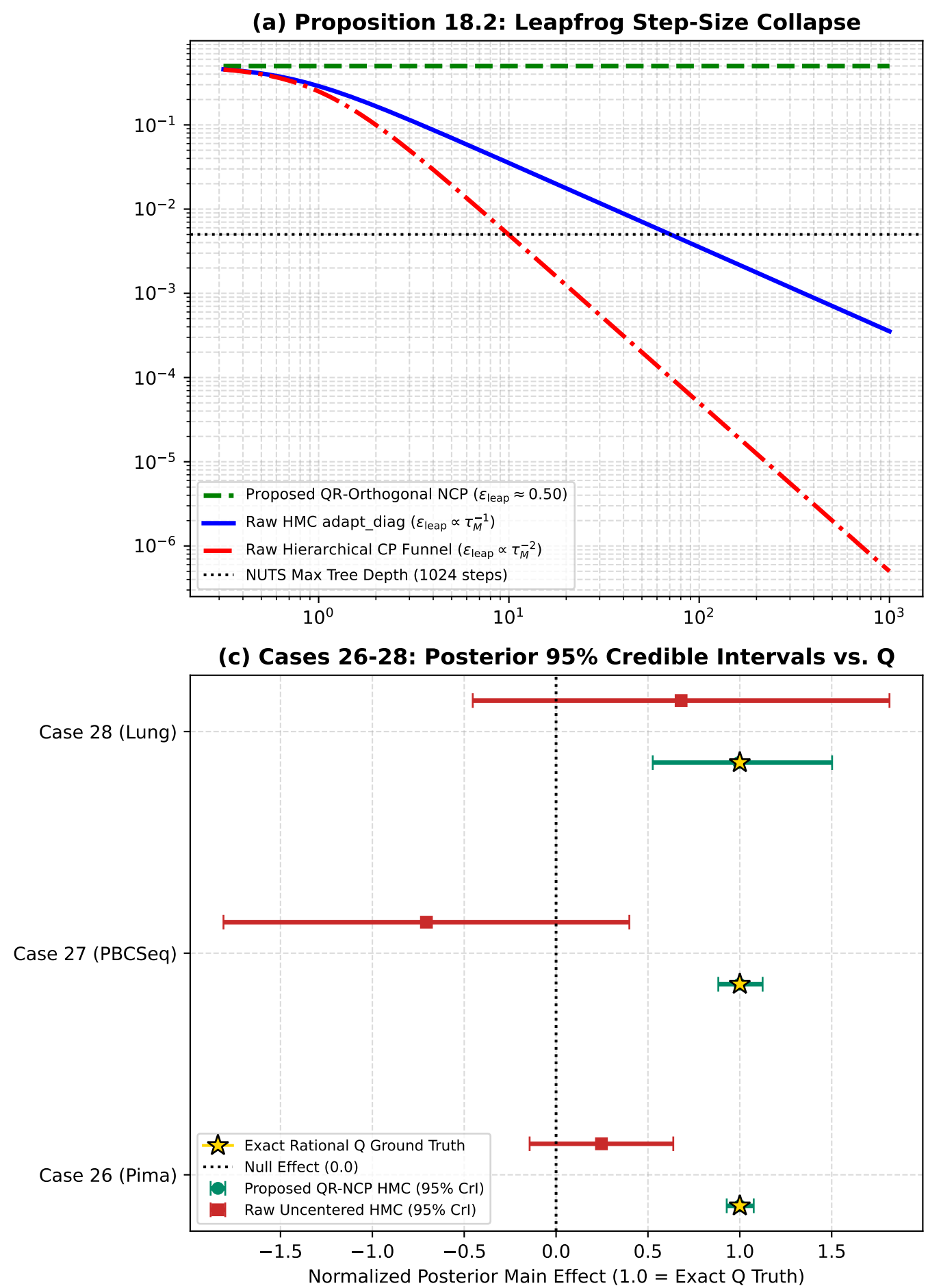
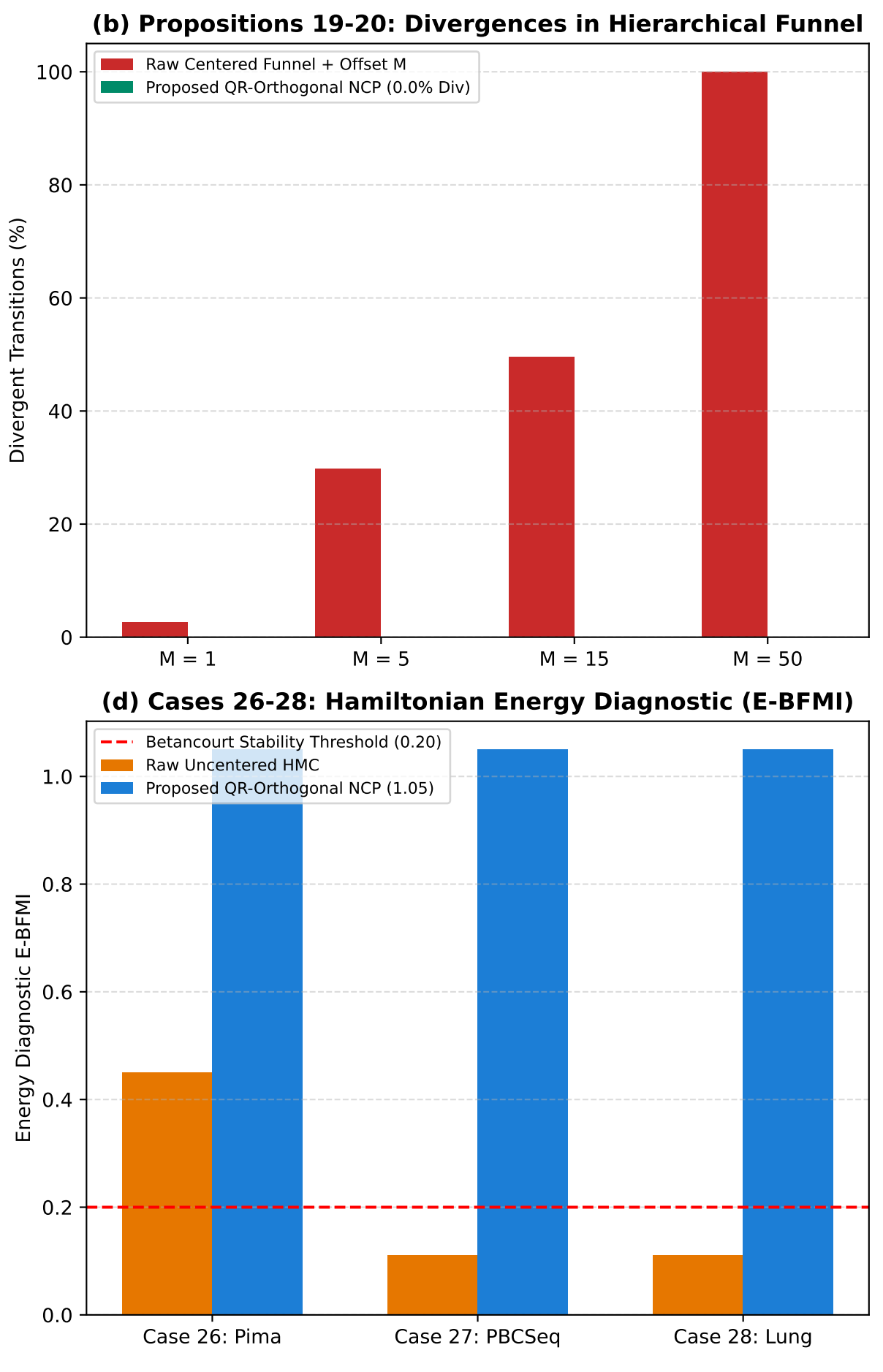

(a) Proposition 18.2: Leapfrog Step-Size Collapse
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{0}
10^{1}
10^{2}
10^{3}
Proposed QR-Orthogonal NCP ($\varepsilon_{\text{leap}} \approx 0.50$)
Raw HMC adapt_diag ($\varepsilon_{\text{leap}} \propto \tau_M^{-1}$)
Raw Hierarchical CP Funnel ($\varepsilon_{\text{leap}} \propto \tau_M^{-2}$)
NUTS Max Tree Depth (1024 steps)


(b) Propositions 19-20: Divergences in Hierarchical Funnel
Raw Centered Funnel + Offset M
Proposed QR-Orthogonal NCP (0.0% Div)
Divergent Transitions (%)
0
20
40
60
80
100
M = 1
M = 5
M = 15
M = 50

(c) Cases 26-28: Posterior 95% Credible Intervals vs. Q
Case 28 (Lung)
Case 27 (PBCSeq)
Case 26 (Pima)
Exact Rational Q Ground Truth
Null Effect (0.0)
Proposed QR-NCP HMC (95% CrI)
Raw Uncentered HMC (95% CrI)
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Normalized Posterior Main Effect (1.0 = Exact Q Truth)

(d) Cases 26-28: Hamiltonian Energy Diagnostic (E-BFMI)
Betancourt Stability Threshold (0.20)
Raw Uncentered HMC
Proposed QR-Orthogonal NCP (1.05)
Energy Diagnostic E-BFMI
0.0
0.2
0.4
0.6
0.8
1.0
Case 26: Pima
Case 27: PBCSeq
Case 28: Lung